BIOCHEMISTRY CUMULATIVE EXAM

October 22, 1994

A newly discovered bacteria has been found to contain a Na^+, K^+ pump that uses the hydrolysis of ATP to pump K^+ into the cell and Na^+ out of the cell to maintain the desired high internal concentration of K^+ and low internal concentration of Na^+. The balanced equation for this reaction is:

$$2\ Na^+_{in} + 2\ K^+_{out} + ATP_{in} \rightarrow 2\ Na^+_{out} + 2\ K^+_{in} + ADP_{in} + Pi_{in}$$

The concentrations inside the cell are [ATP] = 10 mM, [ADP] = 1 mM, [Pi] = 10 mM. The concentrations in the growth media on the outside of the bacterial cell are: $[Na^+_{out}]$ = 100 mM, $[K^+_{out}]$ = 10 mM. The ratios R_1 and R_2 are defined as follows: $R_1 = [Na^+_{in}]/[Na^+_{out}]$, $R_2 = [K^+_{in}]/[K^+_{out}]$. There is no voltage difference between the outside and inside of the cell.

(1) (20) Write an equation for the actual total free energy change ΔG for the complete coupled reaction given above for the pump.

(2) (20) Write an equation for the <u>actual</u> free energy change ΔG_{ATP} for the exergonic component of the reaction involving the hydrolysis of ATP, and calculate the value of ΔG_{ATP} in the bacterial cell under the above conditions.

(2) (20) What bioenergetic condition will be satisfied in order for the pump to produce the minimum value of R_1 and the maximum value of R_2, assuming that no other processes are destroying the Na^+ and K^+ gradients?

(4) (40) Calculate the minimum value of R_1 and the maximum value of R_2 that can be produced under the above conditions. For simplicity, you may assume that $R_1 = 1/R_2$. Also calculate the values of $[Na^+_{in}]$ and $[K^+_{in}]$.

Possible Equations:

$\Delta G^{o'}$(ATP hydrolysis) = - 7.3 KCal/mol.

$\Delta G(X_{out} \rightarrow X_{in}) = RT\ln([X_{in}]/[X_{out}]) + ZF(V_{in} - V_{out})$.

$\Delta G = \Delta G^o + RT\ln([\text{products}]/[\text{reactants}])$.

R = 2 cal/degree.mol F = 23 KCal/volt.mol